

Notes: Farhi, Lerner & Tirole (RAND Journal of Economics, 2013)

Fear of Rejection? Tiered Certification and Transparency

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1 Big picture

- **Question.** Why do many certifiers conceal unsuccessful applications, and how does this opacity shape sellers' certification strategies, buyer information, and the structure of certification markets?
- **Objects of study.** The paper studies certifiers such as journals, rating agencies, lenders, underwriters, universities, licensing exams, standard-setting bodies, and other information intermediaries.
- **Central institutional fact.** Many certifiers do not reveal rejected applications. For example, journals typically do not publicize paper rejections, and rating agencies historically allowed issuers to avoid disclosing unfavorable ratings.
- **Certification hierarchy.** Sellers can apply first to a high-tier certifier and, if rejected, move to a lower-tier certifier. This creates a certification pecking order.
- **Core mechanism.** If buyers infer that a lower-tier certification may reflect earlier rejection by a high-tier certifier, lower-tier certification becomes stigmatized. This stigma makes sellers either avoid ambitious applications or all rush to high-tier certifiers, depending on beliefs.
- **Strategic complementarity.** A seller's choice to try a high-tier certifier makes lower-tier certification look worse, which encourages other sellers to also try the high-tier certifier. Certification choices are strategic complements.
- **Transparency result.** Mandating disclosure of rejections weakly improves sellers' welfare because it removes the hidden externality created by ambitious applications. But it can reduce buyer information because fewer sellers use the high-tier certifier.
- **Certifier market power.** A high-tier monopoly certifier prefers nontransparency because opacity increases the stigma of low-tier certification and therefore increases willingness to pay for high-tier certification.
- **Heterogeneous sellers.** When sellers differ in confidence about quality, nontransparency persists under both competition and monopoly. Transparency is a bad screening device because it hurts lower-confidence types more than high-confidence types.
- **Multitier grading.** Full grading or fine grading is more likely when fine grading is cheap, sellers are impatient, and sellers are less information averse.
- **Economic takeaway.** Opacity about rejection is not simply a bad transparency practice. It is an equilibrium response to seller demand, tiered certification incentives, and the stigma generated by the certification pecking order.

2 Incremental contribution of the paper

2.1 Contribution relative to the literature on certifiers and intermediaries

- Earlier work on certifiers focused mainly on whether certifiers collect information, how they disclose it, and whether they truthfully report quality.
- Farhi, Lerner, and Tirole shift attention to a different but central design choice: whether unsuccessful applications are publicly revealed.
- The paper shows that disclosure policy about rejected applications is not a minor institutional detail. It changes the entire equilibrium structure of certification markets.

Interpretation: The paper expands the theory of certification from *what a certifier says about accepted products* to *what the market infers from the path of failed and successful certification attempts*.

2.2 Contribution relative to models of disclosure and transparency

- Standard transparency arguments usually imply that more disclosure improves information and reduces distortions.
- This paper shows a more subtle tradeoff: transparency can help sellers but make buyers less informed.
- The reason is that transparency changes sellers' application strategies. If disclosure of rejection is mandated, sellers may avoid high-tier certifiers, and high-tier screening is used less often.

Interpretation: The paper contributes a behavioral-equilibrium channel through which mandatory disclosure can reduce information production, even when disclosure eliminates a stigma externality.

2.3 Contribution relative to the literature on market structure

- The paper separates the effect of transparency from the degree of market power in certification.
- Under high-tier certifier monopoly, nontransparency is profitable because it increases the stigma of lower-tier certification.
- Under competition, nontransparency still arises because sellers individually dislike rejection disclosure.
- Thus, opacity is not only a monopoly distortion. It can arise in competitive certification markets as well.

Interpretation: This is an important incremental contribution: the nondisclosure of rejection does not depend solely on certifier market power. It is rooted in seller incentives and equilibrium stigma.

2.4 Contribution relative to rating-agency and journal policy debates

- The paper provides a unified model of practices observed in academic journals, rating agencies, standardized tests, entry examinations, and certification systems.
- It explains why journals tend to use opaque rejection processes and pass-fail publication decisions.
- It explains why rating agencies and exam providers may differ in transparency and grading coarseness.
- It connects policy debates about disclosure of unfavorable ratings to a general theory of certification pecking orders.

Interpretation: The paper's incremental value is partly conceptual: it interprets familiar institutions such as journal rejection confidentiality as equilibrium mechanisms, not just conventions.

2.5 Contribution of the multitier-grading analysis

- The paper endogenizes whether certification is coarse or fine.
- Fine grading is not always optimal even if it reveals more information.
- Sellers may prefer coarse pass-fail certification when they are information averse, whereas fine grading is more attractive when sellers are information loving or when the cost of fine grading is low.

Interpretation: This adds an explicit theory of grading coarseness to a model of certification transparency, linking two institutional choices that are often studied separately.

3 Model primitives and measurement

3.1 Sellers and product quality

- Time is discrete and there is a steady inflow of sellers.
- Each seller has one product whose true quality is initially unknown to sellers and buyers.
- Conditional on the product not being abysmal, quality can be high H or low L .
- Buyer benefits are

$$b_i \in \{b_H, b_L, -\infty\}, \quad b_H > b_L > -\infty.$$

- Conditional on not being abysmal, quality is high with prior probability p and low with probability $1 - p$.
- Buyers do not buy a product unless it is certified to be at least low quality.

Interpretation: The abysmal type gives certification a necessary screening role. Without certification, the product cannot be sold.

3.2 Seller payoff from market beliefs

If buyers hold posterior belief $\hat{p}$ that a certified product is high quality, a seller of true quality $i \in \{H, L\}$ obtains gain

$$S_i(\hat{p}),$$

where $S_i(\hat{p}) > 0$ and is weakly increasing in $\hat{p}$.

Interpretation: The seller's payoff may depend on buyer beliefs because beliefs determine price, demand, matching, adoption, or the seller's broader reputation.

3.3 Certifiers

There are three relevant types of certifiers:

- **Tier-1 certifier:** identifies whether quality is high. It certifies $b = b_H$ or rejects when $b \in \{b_L, -\infty\}$.
- **Tier-2 certifier:** certifies that quality is at least low, namely $b \in \{b_H, b_L\}$, and rejects abysmal quality.
- **Multitier certifier:** reports the exact quality: high, low, or abysmal.

The baseline cost of minimum-standard certification is normalized to zero. Fine grading may require positive cost c .

Interpretation: Tier-1 and tier-2 certifiers correspond to different thresholds. A tier-1 success is more prestigious, but a tier-1 rejection creates a potential stigma.

3.4 Transparency versus opacity

- A certifier is **opaque** if rejected applications are not publicly disclosed.
- A certifier is **transparent** if rejected applications become known to buyers.
- The paper focuses on transparency of the *application path*, not only on the rating outcome.

Interpretation: A product may arrive with tier-2 certification. The crucial inference is whether buyers think this product went directly to a tier-2 certifier or first failed at a tier-1 certifier.

3.5 Timing and seller strategies

A seller can apply to one certifier per period. She chooses when to bring the product to market after certification. If a seller arriving at date t brings the product at date $t + \tau$, and buyer beliefs are $p_{t+\tau}$, seller utility is

$$\delta^\tau S_i(p_{t+\tau}), \quad \delta < 1.$$

The two central strategies are:

- **Safe strategy:** apply directly to a tier-2 certifier.
- **Ambitious strategy:** apply first to a tier-1 certifier and, if rejected, apply to a tier-2 certifier.

Interpretation: The ambitious strategy can generate a high payoff if the seller passes tier-1 certification, but rejection delays market entry and may create stigma.

3.6 Fortuitous disclosure

If a product is delayed because of an earlier rejection, buyers may discover that history with probability $d > 0$. With probability $1 - d$, they do not observe the delay history.

Interpretation: Fortuitous disclosure captures leaks, observable delays, rumors, or other facts that reveal a previous rejection even when certifiers themselves are opaque.

3.7 Seller attitudes toward information

The paper classifies sellers by their preferences over information revelation:

- **Information loving:**

$$pS_H(1) + (1 - p)S_L(0) > pS_H(p) + (1 - p)S_L(p).$$
- **Information averse:**

$$pS_H(1) + (1 - p)S_L(0) < pS_H(p) + (1 - p)S_L(p).$$
- **Information neutral:** equality holds.

Interpretation: This distinction matters for fine grading. Sellers who benefit from matching high-quality products to high-valuation buyers are more information loving; sellers who prefer pooling are more information averse.

4 Equilibrium with minimum-standard certifiers

4.1 Beliefs after tier-2 certification

Let x be the fraction of sellers choosing the ambitious strategy. When buyers see tier-2 certification, their belief is

$$\hat{p} = 0$$

if they know the product arrived late because of a prior rejection. Otherwise,

$$\hat{p} = p(x) = \frac{(1 - x)p}{1 - x + x(1 - p)(1 - d)}.$$

Interpretation: As more sellers choose the ambitious strategy, tier-2 certification becomes more suspicious. The belief $p(x)$ falls as x rises.

4.2 Payoffs from ambitious and safe strategies

If tier-2 certification carries reputation $\hat{p}$, define

$$W^1(\hat{p}) = pS_H(1) + (1-p)\delta[dS_L(0) + (1-d)S_L(\hat{p})], \quad (1)$$

$$W^2(\hat{p}) = pS_H(\hat{p}) + (1-p)S_L(\hat{p}). \quad (2)$$

- W^1 is the payoff from first applying to a tier-1 certifier and then using tier-2 certification after rejection.
- W^2 is the payoff from applying directly to tier-2 certification.

Interpretation: Tier-2 payoffs are more sensitive to reputation than tier-1 payoffs because high-quality sellers pass tier-1 certification and avoid the low-tier stigma.

4.3 Strategic complementarity

The key inequality is

$$\frac{dW^2}{d\hat{p}} \geq \frac{dW^1}{d\hat{p}} \geq 0.$$

Interpretation: When more sellers try tier-1 first, tier-2 certification becomes stigmatized. Because the safe strategy relies entirely on tier-2 reputation, the safe strategy becomes less attractive. Thus one seller's ambitious application increases other sellers' incentives to be ambitious.

4.4 Safe-strategy equilibrium

A safe-strategy equilibrium, $x = 0$, exists if

$$W^2(p) \geq W^1(p),$$

which can be written as

$$(1-p)[(1-\delta)S_L(p) + \delta d(S_L(p) - S_L(0))] \geq p[S_H(1) - S_H(p)].$$

Interpretation: The safe strategy avoids delay and avoids the risk that a low-quality seller is exposed as a tier-1 reject. Its cost is the lost chance to reveal high quality.

4.5 Ambitious-strategy equilibrium

An ambitious-strategy equilibrium, $x = 1$, exists when

$$W^1(0) \geq W^2(0),$$

or

$$pS_H(1) + \delta(1-p)S_L(0) \geq pS_H(0) + (1-p)S_L(0).$$

Interpretation: If everyone first applies to tier-1, then tier-2 certification is very bad news. That stigma can make it individually optimal for every seller to apply first to tier-1.

4.6 Proposition 1

Proposition 1. With minimum-standard certifiers, certification choices are strategic complements. A seller's ambitious strategy encourages other sellers to apply to tier-1 certifiers. In the seller-Pareto-dominant equilibrium:

- sellers use the ambitious strategy if $W^1(p) > W^2(p)$;
- sellers use the safe strategy otherwise;
- ambitious strategies are more likely when fortuitous disclosure is less likely and sellers are more patient.

Interpretation: The model explains why tiered certification can become self-reinforcing. Once low-tier certification becomes stigmatized, even sellers who would prefer not to delay may apply to the high-tier certifier first.

5 Transparency regulation

5.1 Effect of mandated transparency

Under mandated transparency, rejection by a tier-1 certifier is disclosed. The tier-1 payoff becomes

$$W^{1T} = W^1(0),$$

while the payoff from tier-2 certification remains

$$W^{2T}(p) = W^2(p).$$

Interpretation: Transparency makes ambitious strategies less attractive because a seller rejected by tier-1 can no longer hide behind the pool of tier-2 applicants.

5.2 Seller welfare

Transparency can eliminate the ambitious-strategy equilibrium. It weakly raises seller welfare because ambitious strategies impose a stigma externality on safe-strategy sellers.

Proposition 2. Transparency weakly improves sellers' welfare.

Interpretation: Sellers individually may prefer opacity when applying to a high-tier certifier, but collectively opacity can make the lower tier unattractive. Mandated transparency corrects that externality.

5.3 Buyer welfare

Buyer welfare is ambiguous:

- Transparency can reduce information by discouraging high-tier applications.
- But transparency can also clarify the meaning of tier-2 certification.
- In some examples, buyers have no surplus due to price discrimination, so only seller welfare matters.
- In other examples, buyer welfare depends on how beliefs affect matching and pricing.

Interpretation: This is a key policy lesson: disclosure regulation that helps sellers can make buyers less informed about product quality.

5.4 Self-regulation

The paper argues that tier-1 certifiers would not voluntarily agree to disclose rejections, because doing so reduces their collective attractiveness to sellers.

Interpretation: Nontransparency is not necessarily a failure of coordination among certifiers. It may be the individually and collectively preferred policy from the certifiers' perspective.

6 Extensions

6.1 Market power in tier-1 certification

Suppose tier-1 certification is monopolized but tier-2 certification is competitive. The tier-1 monopolist charges a fixed fee F .

Under nontransparency, sellers choose tier-1 first if

$$F < F^{NT} = W^1(p) - W^2(p).$$

Under transparency, sellers choose tier-1 first only if

$$F < F^T = W^1(0) - W^2(p) < F^{NT}.$$

Proposition 3. A monopoly tier-1 certifier has positive market share if and only if condition $W^1(p) > W^2(p)$ is satisfied. When it has market share, it chooses nontransparency to maximize sellers' incentive to apply for tier-1 certification.

Interpretation: The tier-1 monopolist profits from making tier-2 certification look worse. Opacity raises the willingness to pay for tier-1 certification.

6.2 Market power in tier-2 certification

When tier-2 certification is monopolized and tier-1 certification is competitive, tier-2 fees can make direct tier-2 certification less attractive. This can expand the parameter region in which sellers choose ambitious strategies.

Proposition 4. With a monopoly tier-2 certifier, the ambitious-strategy equilibrium occurs for a larger set of parameter values than under competitive tier-2 certification.

Interpretation: Market power in the lower tier makes the lower tier less attractive, indirectly strengthening incentives to try the upper tier first.

6.3 Heterogeneous sellers

Now sellers differ in their prior probability p of being high quality. The seller privately knows p . In equilibrium, there is a cutoff p^* :

- sellers with $p > p^*$ apply first to tier-1;
- sellers with $p < p^*$ apply directly to tier-2.

With monopoly tier-1 certification, the certifier designs a mechanism with a fee $F(p)$ and a disclosure rule. The paper shows that nontransparency is optimal.

Proposition 5. Nontransparency obtains under heterogeneous sellers whether tier-1 certification is competitive or monopolized. With heterogeneous sellers, a monopoly tier-1 certifier charges a type-independent fee.

Interpretation: Transparency is a bad screening tool because it hurts low-confidence sellers more than high-confidence sellers. It therefore reduces total surplus extraction and does not help separate types.

6.4 Multitier certification

A multitier certifier reveals exact quality at cost $c > 0$. It can replace coarse tier-2 certification when fine grading is sufficiently attractive to sellers.

Fine grading is more likely when:

- the cost c is low;
- sellers are impatient, so avoiding multiple certification rounds is valuable;
- sellers are information loving or less information averse.

Proposition 6. Multitier grading occurs for a larger set of parameters when c is lower, when sellers are more impatient, and when sellers are less information averse.

Interpretation: The model explains why some institutions use pass-fail certification while others provide fine grades. The choice depends not only on information technology but also on seller preferences over disclosure.

6.5 Further extensions in the appendix

The appendix studies:

- endogenous quality investment by sellers;
- buyer-side charges by certifiers;
- quick-turnaround certifiers that trade off speed and accuracy.

Interpretation: The appendix shows that the paper’s core mechanisms survive changes in who pays and how quality is produced, while adding new comparative statics such as transparency lowering seller investment quality in some cases.

7 Policy implications and empirical design implications

7.1 Transparency policy

- Mandatory transparency can improve seller welfare by eliminating the hidden rejection externality.
- It may reduce buyer information if sellers avoid high-tier certifiers.
- Hence transparency is not automatically welfare improving for all market participants.

Interpretation: Policy should distinguish between transparency of final certification outcomes and transparency of failed applications.

7.2 Institutional predictions

The model predicts:

- rejection opacity is most common where sellers choose certifiers;
- high-tier certifiers with market power prefer opacity;
- lower-tier certification becomes stigmatized when many sellers move down the hierarchy after rejection;
- fine grading appears when grading costs are low and sellers value detailed information.

Interpretation: These predictions help rationalize the institutional variation summarized in Table 1.

7.3 Empirical applications

Potential empirical settings include:

- academic journal submissions and resubmissions;
- rating shopping in structured finance;
- university admissions and standardized testing policies;
- certification in labor markets, underwriting, organic labeling, and licensing exams.

Interpretation: The paper gives a theory that can be taken to data by observing application histories, resubmission patterns, disclosure rules, and market beliefs about lower-tier certification.

8 Tables, figures, and original equation panels

8.1 Table 1: Transparency and coarseness of grading

Read:

- Table 1 classifies certification environments by two dimensions: whether rejected applications are opaque or transparent and whether grading is coarse or fine.
- Scientific journals, underwriting, job markets, and university admissions are mostly opaque.
- Some entry-level examinations and medical licensing exams are transparent.
- Rating agencies and entry-level examinations provide finer grading than most journals.

Economic message: The paper’s theory is designed to explain both dimensions: why rejection opacity is common and why some certifiers provide fine grades while others use minimum standards.

TABLE 1 Transparency and Coarseness of Grading

		Opaque	Transparent
Coarseness: grading strategy is	minimum standard;	• (most) scientific journals • job market • underwriting • university admissions • bepress* • rating agencies	• (many) state medical licensing exams • (some) entry-level examinations (bar exam, series 7)
	fine;		• (some) entry-level examinations (SAT, ...)

This table summarizes the transparency and coarseness of grading of different certification environments.

*bepress, which was founded in 1999 by Berkeley faculty members was, until acquired by commercial publisher De Gruyter in 2011, a scholarly publishing service with many open access-like features and quality ratings (usually four different tiers).

... not reveal rejected applications. By contrast, some entry-level examinations (SAT, GMAT, ...)

8.2 Original payoff equations

Read:

- The original panel shows beliefs after tier-2 certification, the payoff from tier-1-first certification, and the payoff from direct tier-2 certification.
- The key mathematical feature is that tier-2 payoffs are more sensitive to the reputation of tier-2 certification than tier-1 payoffs.

Economic message: These equations generate the strategic complementarity in certification choices.

tifier, and apply to a tier-2 certifier in case of rejection). Fraction $1 - x$ select the *safe strategy* directly to a tier-2 certifier).

When faced with a product certified by a tier-2 certifier, buyers form beliefs:

$$\hat{\rho} = \begin{cases} 0 & \text{if they know the product introduction is delayed (as they then infer a rejection in the previous period), and} \\ \hat{\rho} = \hat{\rho}(x) \equiv (1-x)\rho / [1-x + x(1-\rho)(1-d)] & \text{otherwise.} \end{cases}$$

ote that $\hat{\rho}(x)$ decreases from ρ to 0 as x increases from 0 to 1.

Let

$$W^1(\hat{\rho}) \equiv \rho S_H(1) + (1-\rho)\delta[S_L(0) + (1-d)S_L(\hat{\rho})]$$

and

$$W^2(\hat{\rho}) \equiv \rho S_H(\hat{\rho}) + (1-\rho)S_L(\hat{\rho})$$

note the expected payoffs¹⁴ when applying to a tier-1 or tier-2 certifier, when certification a tier-2 certifier delivers reputation $\hat{\rho}$. A crucial property is that, due to (i) discounting, (ii) fortuitous disclosure, and (iii) the fact that high types pass tier-1 certification, the payoff from tier-2 certification is more sensitive to the perception of quality following tier-2 certification than the payoff from tier-1 certification:

$$\frac{\partial W^2}{\partial \hat{\rho}} \geq \frac{\partial W^1}{\partial \hat{\rho}} \geq 0.$$

Safe-strategy equilibrium. It is an equilibrium for sellers to all adopt a safe strategy ($x = 0$)¹⁵ if $W^2(\rho) \geq W^1(\rho)$:

$$\rho S_H(\rho) + (1-\rho)S_L(\rho) > \rho S_H(1) + \delta(1-\rho)[(1-d)S_L(\rho) + dS_L(0)].$$

Beliefs and payoffs after tier-2 certification

or

$$(1-\rho)[(1-\delta)S_L(\rho) + \delta d[S_L(\rho) - S_L(0)]] \geq \rho[S_H(1) - S_H(\rho)]. \quad (1)$$

Condition (1) captures the costs and benefits of a safe strategy. A safe strategy avoids delaying introduction when quality is low, thereby economizing $(1-\delta)S_L(\rho)$. It also prevents the stigma associated with fortuitous disclosure and thereby provides gain $\delta d[S_L(\rho) - S_L(0)]$. The cost of a safe strategy is, of course, the lack of recognition of a high quality and the concomitant loss $S_H(1) - S_H(\rho)$.

Insurprisingly, the more likely a safe-strategy equilibrium is to emerge, the lower the discount factor (e.g., the longer the certification length), and the higher the rate of fortuitous disclosure. Indeed, when $\delta = 1$, the safe-strategy equilibrium never exists (i.e., even for $d = 1$) if the seller is information loving.

Ambitious-strategy equilibrium. Next, consider an equilibrium in which all sellers adopt an ambitious strategy. Certification by a second-tier certifier is then very bad news. Thus, $x = 1$ is an equilibrium if and only if $W^1(0) \geq W^2(0)$:

$$\rho S_H(1) + \delta(1-\rho)S_L(0) \geq \rho S_H(0) + (1-\rho)S_L(0). \quad (2)$$

¹⁴ Conditional on $b \in [b_L, b_H]$.

¹⁵ Observed certifier rankings always start with "tier-1," almost by definition. One interpretation of our model is this particular class of sellers (products) is an identifiable subgroup of sellers (products), who in equilibrium apply tier-2 certifiers (on this, see also the second subsection of Section 4 below). Another interpretation speaks to the very notion of "tier-1," "tier-2," etc. What we here call "tier-2" could in practice be called "tier-1" if no seller applied that we define as "tier-1" certifiers. For example, no "super tier-1" journal has been created that would be more

Safe and ambitious strategy conditions

8.3 Figure 1: Submission strategies' payoffs

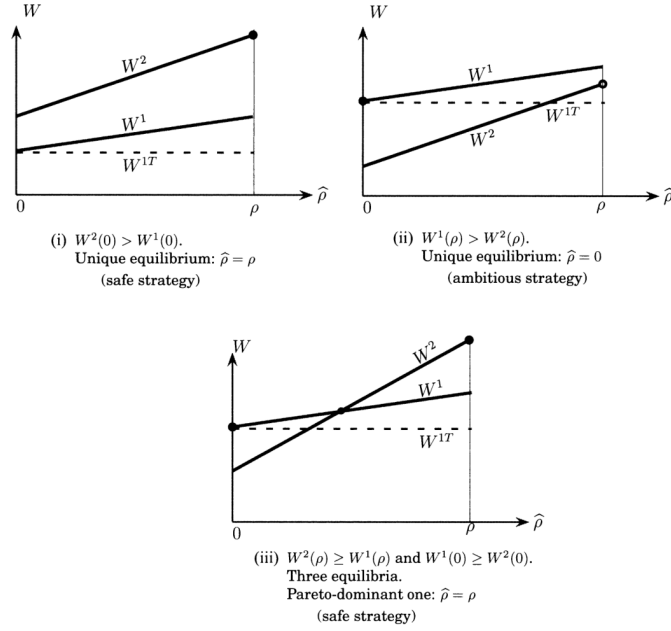
Read:

- Panel (i) shows a unique safe-strategy equilibrium: direct tier-2 application dominates.
- Panel (ii) shows a unique ambitious-strategy equilibrium: tier-1-first application dominates.
- Panel (iii) shows multiple equilibria due to strategic complementarities.
- The dotted line corresponds to the transparency regime discussed in Proposition 2.

Economic message: Figure 1 is the visual summary of the model. Certification pecking orders can create equilibrium stigma and multiple self-fulfilling application patterns.

FIGURE 1

SUBMISSION STRATEGIES' PAYOFFS



This figure depicts the payoffs from the safe and ambitious submission strategies as a function of the average quality of applications to Tier-2 certifiers.

An example of impatient sellers in many American universities are junior faculty members.

8.4 Original transparency and multitier-certification panels

Read:

- The transparency panel shows how mandated disclosure replaces $W^1(p)$ with $W^1(0)$ and can eliminate the ambitious equilibrium.
- The multitier panel shows the conditions under which full grading displaces tiered minimum-standard certification.

Economic message: Transparency and fine grading are distinct institutional tools. Transparency changes the stigma of rejection; fine grading changes the precision of final quality disclosure.

Suppose that a regulator can enforce transparency of applications (this amounts to setting $\hat{\rho} = 1$). For given beliefs $\hat{\rho}$ attached to tier-2 certification, a certification by a tier-2 certifier still holds (“T” refers to “transparency”) $W^{2T}(\hat{\rho}) = W^2(\hat{\rho})$.

By contrast, application to a tier-1 certifier yields a lower payoff than in the absence of transparency:

$$W^{1T} = W^1(0) < W^1(\hat{\rho}) \quad \text{whenever } \hat{\rho} > 0.$$

The analysis of the first subsection of Section 3 applies, with $W^1(\hat{\rho})$ replaced by $W^1(0)$. Application to a transparent tier-1 certifier (with payoffs as depicted by the dashed horizontal line in Figure 1) is an equilibrium behavior if and only if

$$W^1(0) \geq W^2(\rho).$$

if $W^1(0) < W^2(\rho) < W^1(\rho)$, or

$$\begin{aligned} \rho S_H(1) + \delta(1 - \rho)S_L(0) &< \rho S_H(\rho) + (1 - \rho)S_L(\rho) < \rho S_H(1) \\ &+ \delta(1 - \rho)[(1 - d)S_L(\rho) + dS_L(0)], \end{aligned}$$

transparency requirement increases the sellers’ welfare; see case (ii) in Figure 1, where $\hat{\rho} = \rho$ is not an equilibrium under opaque applications and becomes (a Pareto-dominant) one under transparency. In the other parameter configurations (cases (i) and (iii) in Figure 1), transparency has no impact on equilibrium outcome and welfare.

Proposition 2 *Transparency weakly improves sellers’ welfare.* Intuitively, transparency makes ambitious strategies less appealing to individual sellers. It thereby eliminates the basic externality associated with ambitious strategies, which depreciate tier-2 certification. Proposition 2 therefore is a natural result.

Self-regulation. Relatedly, would tier-1 certifiers agree among each other not to compete on the

Transparency regulation and Proposition 2

Second, consider a safe-strategy equilibrium ($x = 0$), and so condition (1) obtains. This equilibrium is robust to the introduction of full grading if and only if furthermore

$$\rho S_H(\rho) + (1 - \rho)S_L(\rho) \geq \rho S_H(1) + (1 - \rho)S_L(0) - c$$

$$\iff c \geq \bar{c} \equiv \bar{c} \equiv \rho[S_H(1) - S_H(\rho)] - (1 - \rho)[S_L(\rho) - S_L(0)].$$

Note that when $c = 0$, this condition holds if and only if the sellers are information averse.²¹

To sum up, sellers resort to multitier grading when its cost c is low, when sellers are impatient (δ is low), and when sellers are information neutral or loving.

Proposition 6. Multitier grading occurs for a larger set of parameters, the lower its cost c , the more impatient the sellers (the lower δ), and the less information averse the sellers are (the higher $\rho[S_H(1) - S_H(\rho)] - (1 - \rho)[S_L(\rho) - S_L(0)]$).

Entry attempts into multitier grading seem to resonate with academic experience. Illustrations include fine grading by prepress and the proliferation of prizes offered by tier-2 journals and not by tier-1 journals). Alternatively, multitier grading can piggyback on an existing tier-1 certifier. For example, the American Economic Association launched the *American Economic Journals*; to be certain, the AEA is careful about not describing the *AEJs* as lower tier relative to the *American Economic Review*, and the process is a bit different from a multitier process, but the fundamentals driving the introduction of the *AEJs* are the same as those developed in this section. *Nature* operates a very similar system via its set of “Nature Research Journals,” of which *Nature Biotechnology* is probably one of the best known.²²

Proposition 6 may also shed some light on rating agencies’ practice of fine grading. As we observed in Example 3 (Section 2), bond ratings not only certify the quality of an issue

Multitier grading and Proposition 6

9 Short summary

- The paper studies certification markets where sellers choose certifiers and may apply sequentially.
- Rejected applications are often not disclosed because sellers dislike stigma, and certifiers benefit from preserving the stigma of lower-tier certification.
- Tiered certification creates strategic complementarities: if many sellers first apply to high-tier certifiers, lower-tier certification looks bad.
- With minimum-standard certifiers, equilibria can involve direct lower-tier certification, tiered certification, or multiple equilibria.
- Mandated transparency weakly improves seller welfare by reducing the externality created by hidden rejections.
- Buyer welfare is ambiguous because transparency can reduce the amount of high-tier screening.
- Monopoly high-tier certifiers prefer opacity because it increases willingness to pay for high-tier certification.
- Nontransparency also arises with heterogeneous sellers because transparency is a poor screening instrument.
- Multitier grading is more likely when it is cheap, when sellers are impatient, and when sellers are less information averse.
- The model explains why many certification markets combine rejection opacity with coarse grading.

10 Conclusion

10.1 Core conclusion

Farhi, Lerner, and Tirole show that opacity about rejected applications can arise naturally in certification markets. Sellers individually dislike having rejections disclosed, and high-tier certifiers may prefer opacity because it increases the stigma attached to lower-tier certification. The resulting certification hierarchy can generate strategic complementarities and multiple equilibria.

10.2 Economic intuition

The central intuition is that lower-tier certification is interpreted partly through the seller's application path. If buyers believe that many lower-tier-certified products were previously rejected by a higher-tier certifier, then lower-tier certification is stigmatized. This makes lower-tier certification less attractive, which encourages more sellers to apply first to the higher tier, validating the stigma.

10.3 Incremental takeaway

The paper's incremental contribution is to connect rejection confidentiality, tiered certification, certifier market power, and grading coarseness in one framework. It shows that opacity can survive competition, that transparency can help sellers but make buyers less informed, and that fine grading appears only under specific cost and preference conditions.

10.4 Takeaway for reading other certification papers

When evaluating certification institutions, the relevant design variables are not only rating accuracy or truthfulness. One must also ask who observes failed applications, how sellers move across certification tiers, whether lower-tier certification is stigmatized, and whether the grading system is coarse or fine.